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Week 4, Lecture 1
Quantum optical information

Reference: Nielsen & Chuang ch. 1.2, 1.3

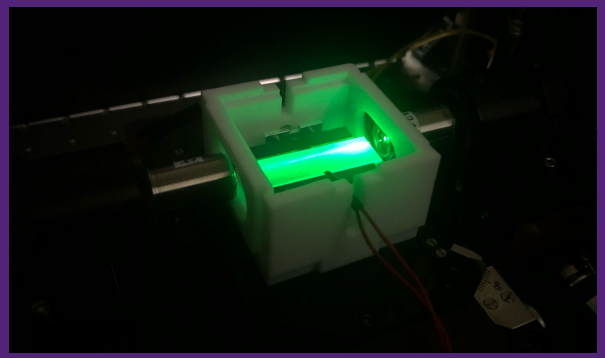
Who am I?



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Research: experimental quantum optics





Originally from Bari (Italy)



PhD in Paderborn (Germany)

What we will cover today…

01

Quantum information

02

Photons as qubits

03

Entanglement

Slides by Dr Josephine Dias



01

Quantum information

Qubits

A qubit is a two-level quantum system.

'computational basis' $\{|0\rangle, |1\rangle\}$

A pure qubit state has the general form:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

α, β are complex numbers called *amplitudes*

$$|\alpha|^2 + |\beta|^2 = 1 \qquad |\alpha|^2 = \alpha \times \alpha^*$$

global phase is irrelevant

$$|\psi\rangle = e^{i\theta}(\alpha|0\rangle + \beta|1\rangle)$$

If the qubit is measured in the computational basis:

- We find result $|0\rangle$ with probability $|\alpha|^2$
- We find result $|1\rangle$ with probability $|\beta|^2$

"braket notation"

Qubits as vectors

- Kets are represented by a 2×1 column vector

$$|0\rangle = \begin{bmatrix}1\\0\end{bmatrix} \quad |1\rangle = \begin{bmatrix}0\\1\end{bmatrix}$$

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle \quad \Rightarrow \quad |\psi\rangle = \begin{bmatrix}\alpha\\\beta\end{bmatrix}$$

- Bras are represented by a 1×2 row vector $\quad \langle 0| = \begin{bmatrix}1 & 0\end{bmatrix} \quad \langle 1| = \begin{bmatrix}0 & 1\end{bmatrix}$

$$\langle\psi| = \alpha^*\,\langle 0| + \beta^*\,\langle 1| \quad \Rightarrow \quad \langle\psi| = \begin{bmatrix}\alpha^* & \beta^*\end{bmatrix}$$

$$\langle\psi|\psi\rangle = \begin{bmatrix}\alpha^* & \beta^*\end{bmatrix}\begin{bmatrix}\alpha\\\beta\end{bmatrix} = |\alpha|^2 + |\beta|^2 = 1$$

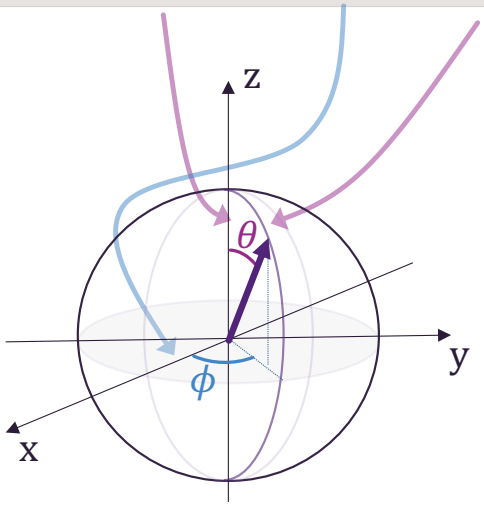
The Bloch sphere

- Because the global phase is irrelevant, we can choose one coefficient to be real, eg. α.
- With the normalisation condition $|\alpha|^2 + |\beta|^2 = 1$, this allows us to parametrise our qubit with two angles: θ and ϕ.

$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$ can be rewritten as

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

$$0 \leq \phi \leq 2\pi$$
$$0 \leq \theta \leq \pi$$



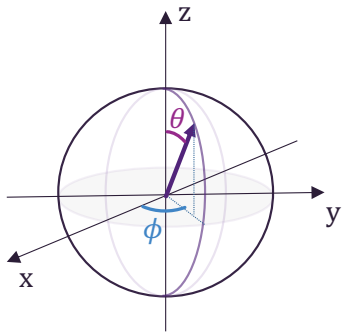
θ, ϕ define a point on the surface of a three-dimensional unit sphere known as the **Bloch sphere**

The Bloch sphere – Computational basis states

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

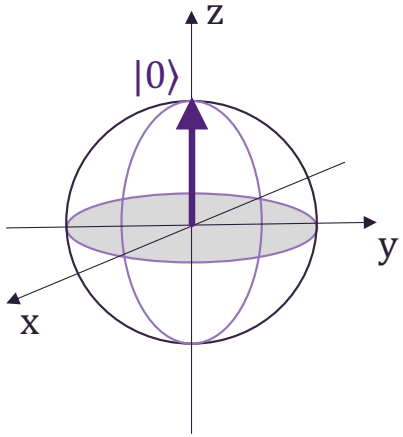
$0 \leq \phi \leq 2\pi$

$0 \leq \theta \leq \pi$

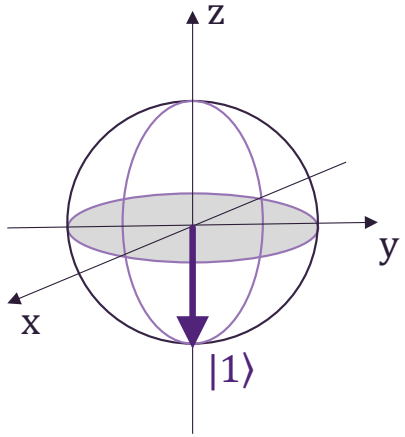


global phase is irrelevant

$\theta = 0$ ➡ $|\psi\rangle = |0\rangle$



$\theta = \pi$ ➡ $|\psi\rangle = \cancel{e^{i\phi}}|1\rangle = |1\rangle$



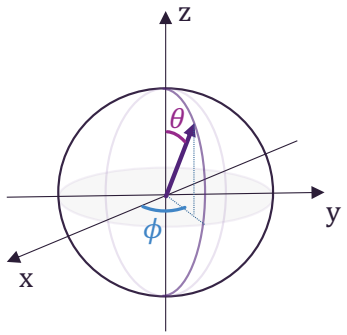
orthogonal states are antiparallel

The Bloch sphere – $|+\rangle$ and $|-\rangle$

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

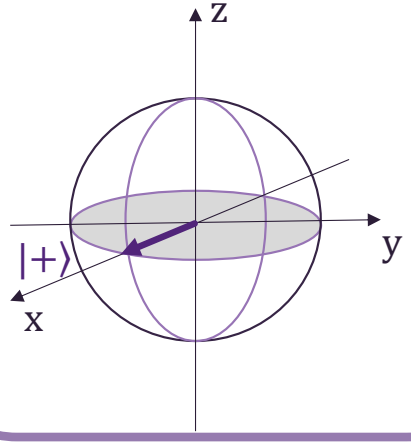
$0 \leq \phi \leq 2\pi$

$0 \leq \theta \leq \pi$



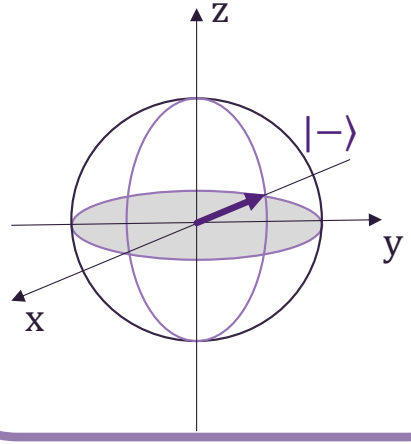
$\theta = \pi/2 \quad \phi = 0$

$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$



$\theta = \pi/2 \quad \phi = \pi$

$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle$



$|+\rangle$ and $|-\rangle$ are orthogonal
$\Rightarrow$antiparallel on the Bloch sphere

Qubit operations – Pauli matrices

The Pauli matrices represent an important group of qubit operations to remember

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \qquad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \qquad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

also called $\sigma_x, \sigma_y, \sigma_z$

Each Pauli matrix is its own inverse

$$X^2 = Y^2 = Z^2 = I$$

Cyclic commutation

$$[X, Y] = 2iZ$$
$$[Y, Z] = 2iX$$
$$[Z, X] = 2iY$$

Qubit operations – eigenvectors of Pauli matrices

Each pauli matrix has eigenvalues +1 and -1

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

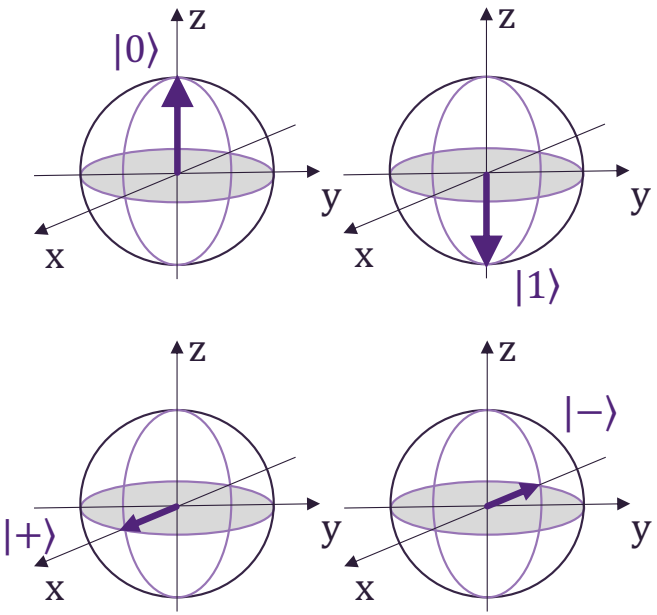
Eigenvectors of Z

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$ along z axis

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Eigenvectors of X

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$
$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$ along x axis



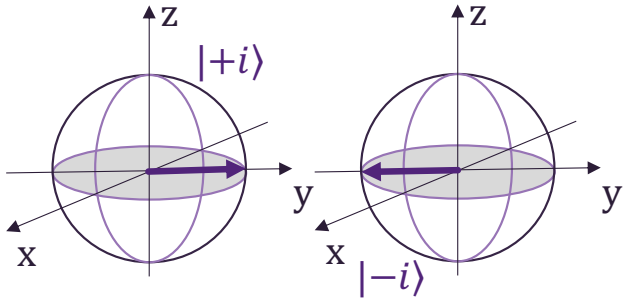
Qubit operations – eigenvectors of Pauli matrices

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

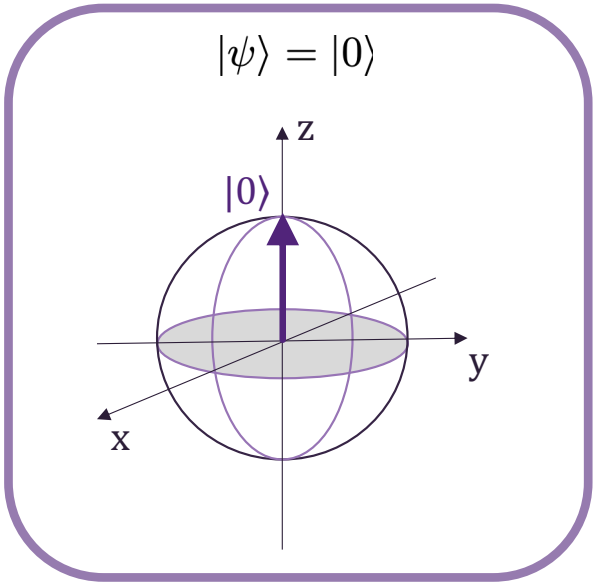
Eigenvectors of Y

$$|+i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$$
$$|-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$$

} along y axis



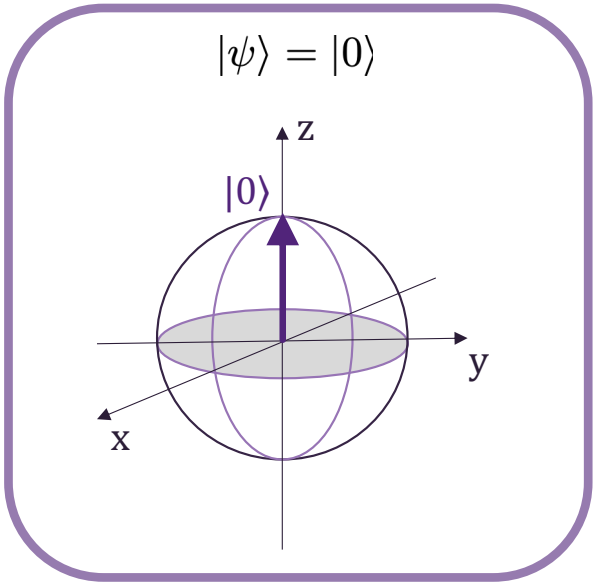
Qubit operations – on the Bloch sphere



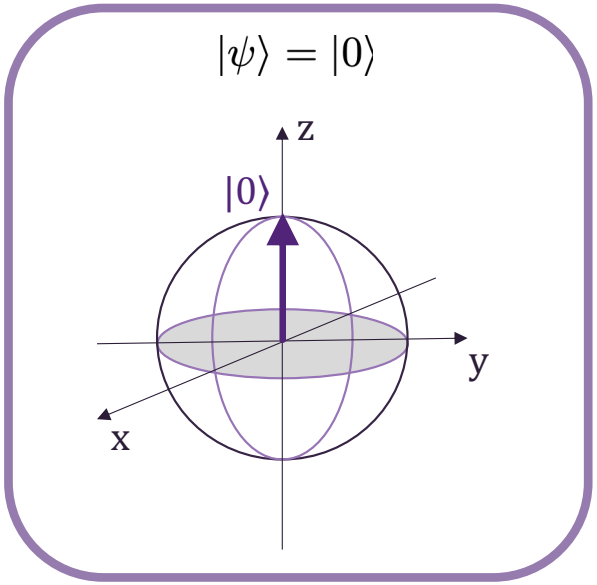
Apply

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$Z|0\rangle = |0\rangle$$
$$Z|1\rangle = -|1\rangle$$



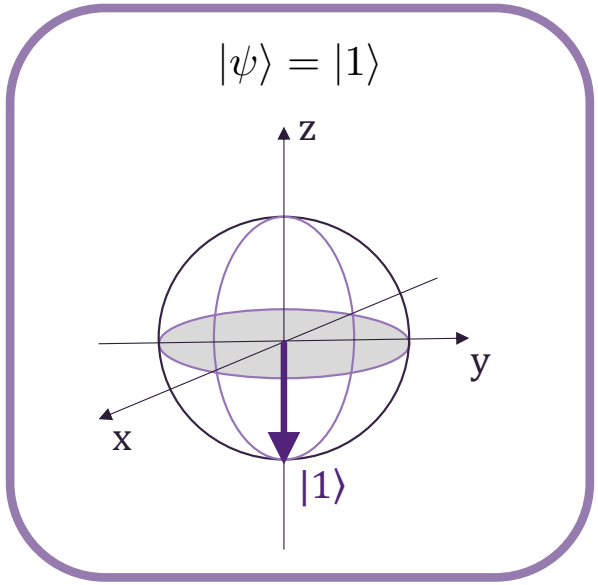
Qubit operations – on the Bloch sphere



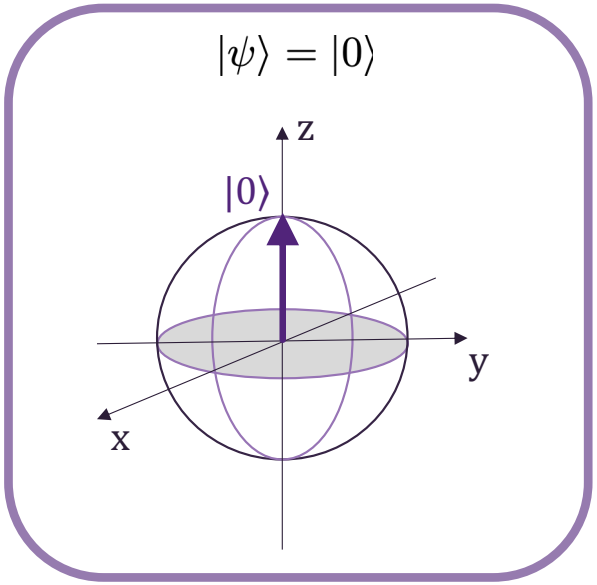
Apply

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$X|0\rangle = |1\rangle$$
$$X|1\rangle = |0\rangle$$



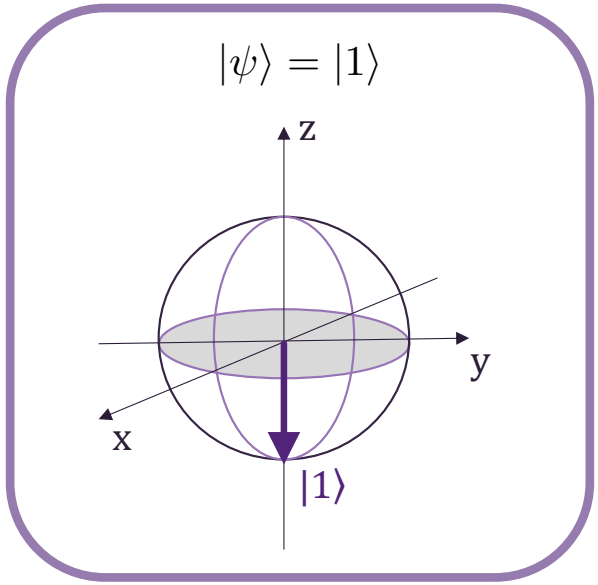
Qubit operations – on the Bloch sphere



Apply

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$Y|0\rangle = i|1\rangle$$
$$Y|1\rangle = -i|0\rangle$$



The Pauli X, Y and Z matrices apply a π rotation about the corresponding axis to a qubit on the Bloch sphere.

Qubit operations - Rotations

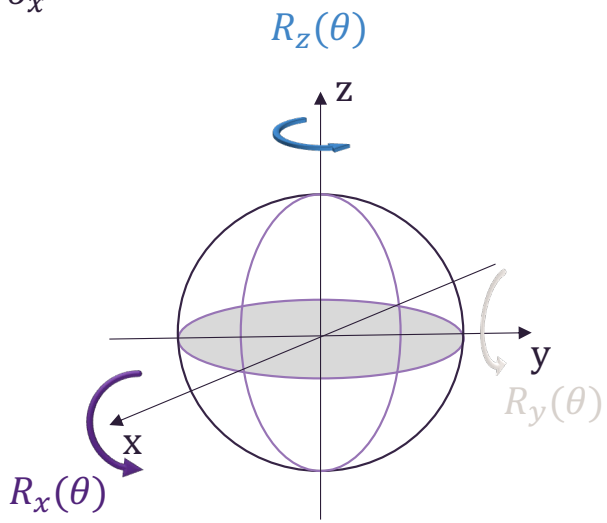
Matrix expontentials of the Pauli matrices give us the rotation operators

$$R_x(\theta) = e^{-i\theta X/2} \qquad X = \sigma_x$$
$$= \cos(\theta/2)\, I - i \sin(\theta/2)\, \sigma_x$$
$$= \begin{bmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$$

Each operator is a rotation about the x,y or z axis of the Bloch sphere

$$R_y(\theta) = e^{-i\theta Y/2}$$
$$= \cos(\theta/2)\, I - i \sin(\theta/2)\, \sigma_y$$
$$= \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$$

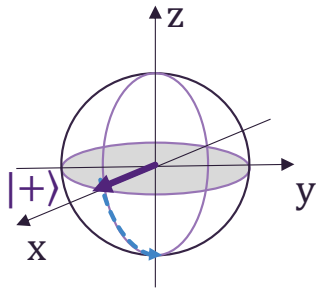
$$R_z(\theta) = e^{-i\theta Z/2}$$
$$= \cos(\theta/2)\, I - i \sin(\theta/2)\, \sigma_z$$
$$= \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}$$



Qubit operations - Rotations

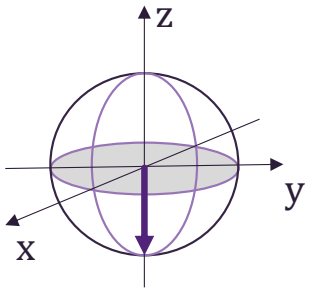
Example 1: $\pi/2$ rotation around y axis

$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$



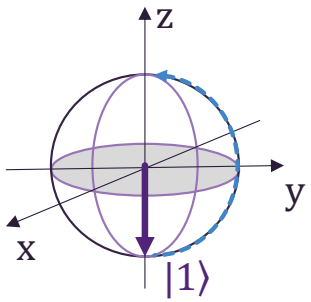
Apply $R_y(\pi/2)$

$$\frac{1}{\sqrt{2}}\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix}\begin{bmatrix}\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}}\end{bmatrix} = \begin{bmatrix}0\\ 1\end{bmatrix}$$



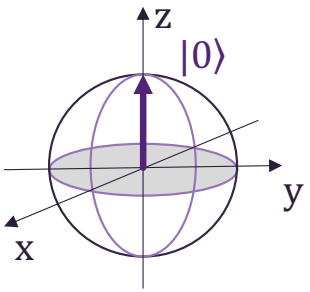
$|1\rangle$

Example 2: π rotation around x axis



Apply $R_x(\pi)$

$$\begin{bmatrix}0 & -i\\ -i & 0\end{bmatrix}\begin{bmatrix}0\\ 1\end{bmatrix} = \begin{bmatrix}1\\ 0\end{bmatrix}$$



$R_y(\pi/2)$ followed by $R_x(\pi)$ gives us the Hadamard gate

$$H = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$$

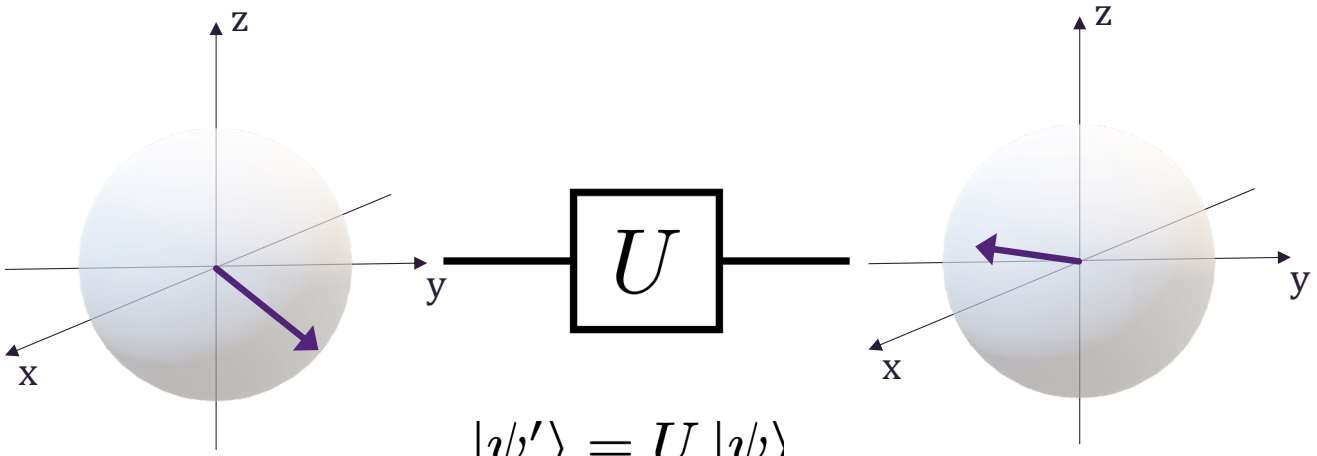
$$R_x(\pi)R_y(\pi/2) = -iH$$

Circuit notation

Examples of single qubit operations and their quantum circuit notation

Pauli-x		$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-y		$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-z		$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard		$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

Unitary operations



$$|\psi'\rangle = U\,|\psi\rangle$$

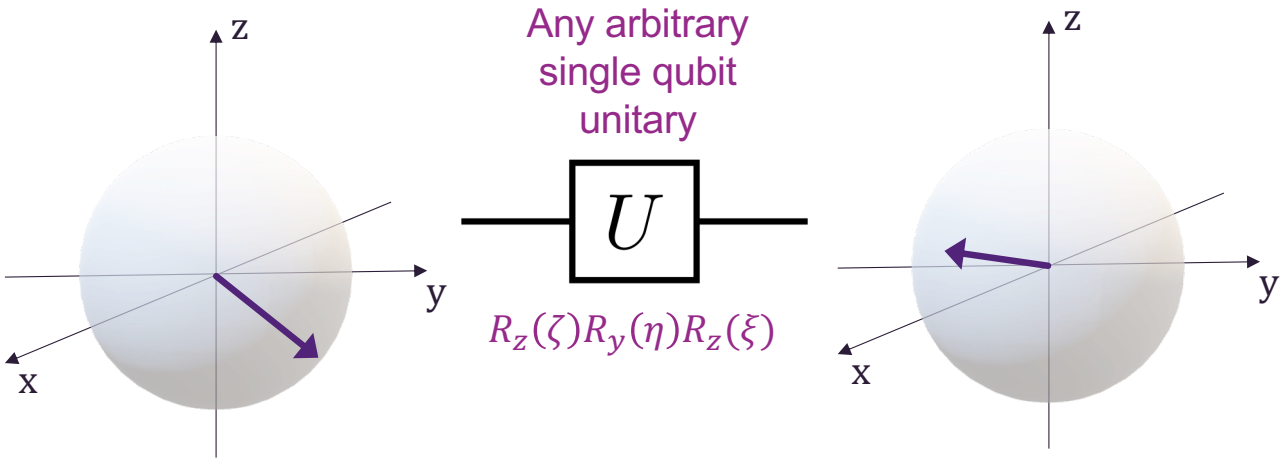
2×2 unitary matrix

$$U^\dagger = U^{-1}$$

$$U^\dagger U = I$$

- Valid single qubit operations are represented by 2×2 unitary matrices
- Unitarity means $U^\dagger = U^{-1}$
- Unitary transformations preserves the norm of the qubit

Decomposing single qubit unitaries

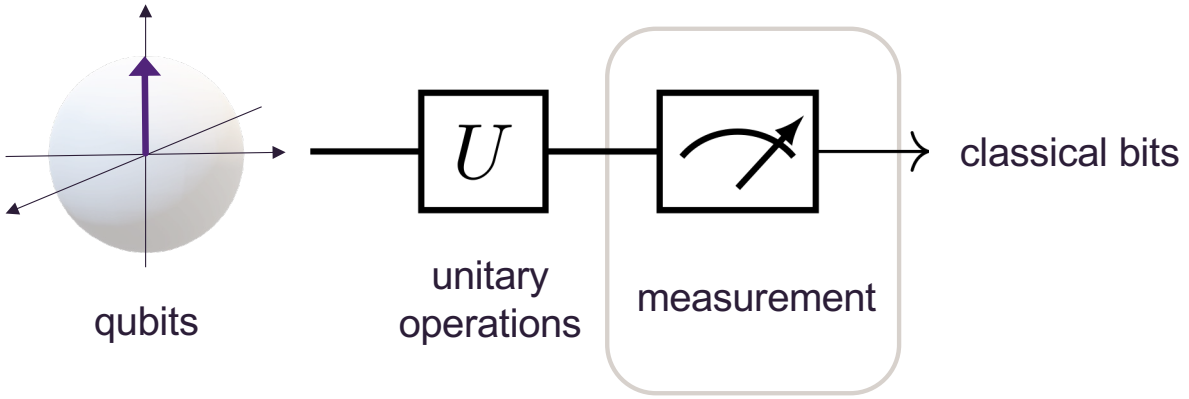


Any arbitrary single qubit unitary operation can be decomposed into rotations

$$U = \begin{bmatrix} e^{-i\zeta/2} & 0 \\ 0 & e^{i\zeta/2} \end{bmatrix} \begin{bmatrix} \cos(\eta/2) & -\sin(\eta/2) \\ \sin(\eta/2) & \cos(\eta/2) \end{bmatrix} \begin{bmatrix} e^{-i\xi/2} & 0 \\ 0 & e^{i\xi/2} \end{bmatrix}$$

$$\zeta, \eta, \xi \in \mathbb{R}$$

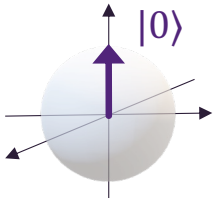
Qubit measurement

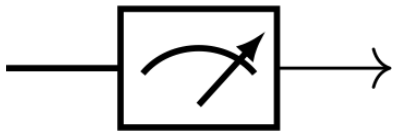


$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ → measurement in computational basis → $|0\rangle$ or $|1\rangle$

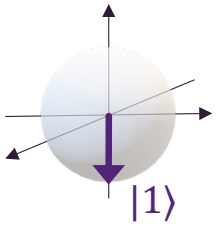
Qubit measurement – Computational basis

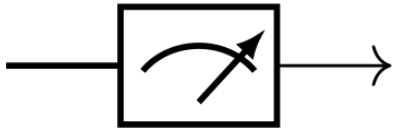
measure in the computational basis



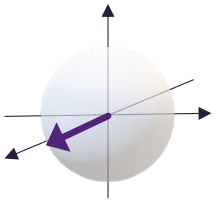


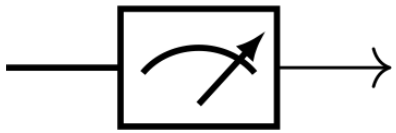
$|0\rangle$ with certainty





$|1\rangle$ with certainty





50% chance for $|0\rangle$
50% chance for $|1\rangle$

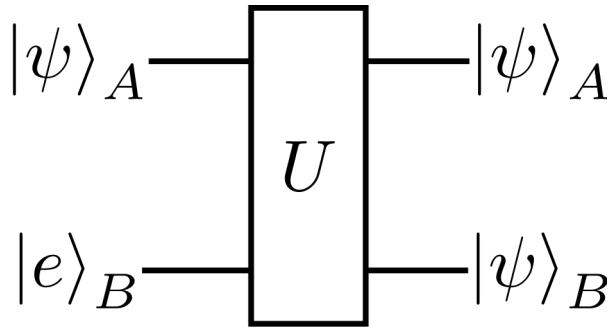
$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Only qubits that are already in an eigenstate of the observable being measured will have measurement outcomes predictible with certainty, with other qubits we can only calculate the probability of each measurement outcome

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Can we copy a quantum state?

Given a quantum state $|\psi\rangle$ can we write down a quantum circuit that creates two copies of $|\psi\rangle$?



Can we copy a quantum state?

$$U\left[|\psi\rangle_A |e\rangle_B\right] = |\psi\rangle_A |\psi\rangle_B \qquad U\left[|\phi\rangle_A |e\rangle_B\right] = |\phi\rangle_A |\phi\rangle_B$$

$$\langle e|_B \langle \phi|_A \underbrace{U^\dagger U}_{I} |\psi\rangle_A |e\rangle_B = \langle \phi|_A \langle \phi|_B |\psi\rangle_A |\psi\rangle_B$$

assume $|e\rangle$ is normalised

$$\langle \phi|\psi\rangle_A \underbrace{\langle e|e\rangle_B}_{1} = \langle \phi|\psi\rangle_A \langle \phi|\psi\rangle_B$$

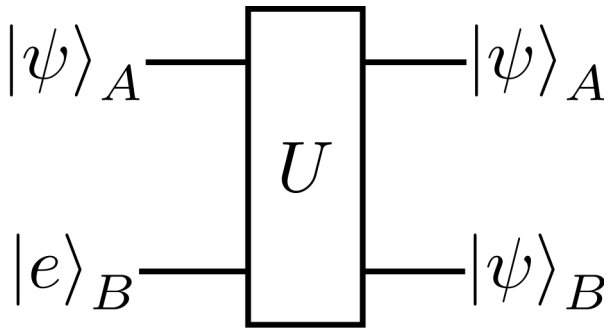
$$\langle \phi|\psi\rangle_A = \left(\langle \phi|\psi\rangle_A\right)^2$$

which is an equation of the form $x = x^2$

only solutions are:

$\langle \phi|\psi\rangle = 0$ → $|\phi\rangle$ and $|\psi\rangle$ are orthogonal

$\langle \phi|\psi\rangle = 1$ → $|\phi\rangle$ and $|\psi\rangle$ are the same



→ no cloning for arbitrary quantum states

No-cloning theorem

The **No-cloning theorem** states that there is no unitary operator U on $H \otimes H$ such that for arbitrary normalised $|\psi\rangle_A$ and $|e\rangle_B$

$$U(|\psi\rangle_A|e\rangle_B) = |\psi\rangle_A|\psi\rangle_B$$

up to some global phase

→ it is impossible to create an identical copy of an unknown quantum state

$$U\left[|\psi\rangle_A |e\rangle_B\right] = |\psi\rangle_A |\psi\rangle_B$$

impossible

Some notes on the No-cloning theorem…

- A known operation that generates some state $|\psi\rangle$ can just be repeated to get more copies of $|\psi\rangle$
- But if you were given an unknown quantum state, there is no unitary operator that can copy or clone that quantum state
- The no cloning theorem has many important consequences in quantum information
- Classical error correction techniques won't work – need quantum error correction
- Classical information can be copied but quantum information can't be copied

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02

Photons as qubits

Light has many degrees of freedom

Week 1:

From one mode to many modes

A general electromagnetic field contains many modes:

$$\mathbf{A}(\mathbf{r},t) = \sum_j q_j(t)\mathbf{u}_j(\mathbf{r})$$

Each mode j has its own:

- Spatial profile $\mathbf{u}_j(\mathbf{r})$
- Angular frequency ω_j
- Polarisation
- Time-dependent amplitude $q_j(t)$

The modes are orthonormal:

$$\int_V \varepsilon_0 \mathbf{u}_j(\mathbf{r}) \cdot \mathbf{u}_k(\mathbf{r})\, \mathrm{d}^3\mathbf{r} = \delta_{jk}$$

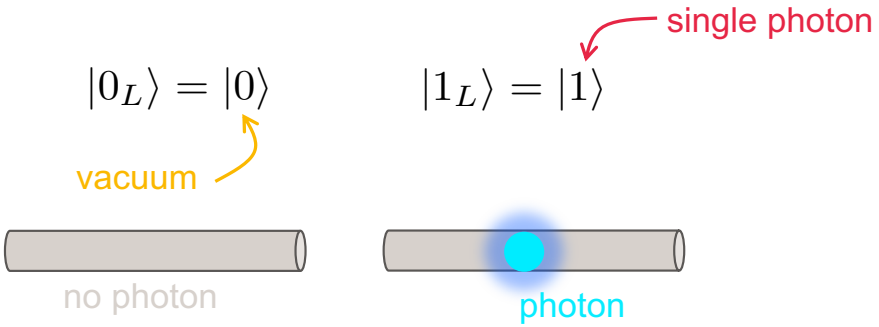
All these different degrees of freedom of photons can contain quantum information

Photons as qubits

How can we encode a qubit using photons?

$$|\psi\rangle = \alpha\,|0_L\rangle + \beta\,|1_L\rangle$$

Single-Rail Encoding

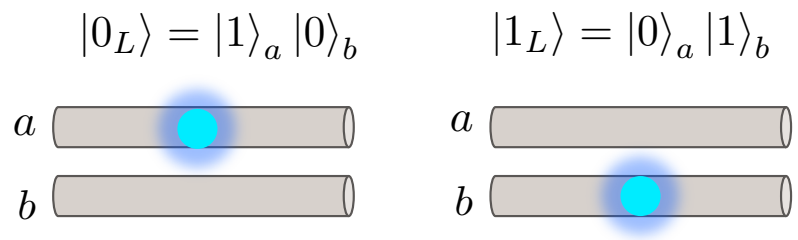


Use one single mode: information is encoded in the **presence** or **absence** of a photon

Challenge:

Photon loss: maps $|1\rangle \rightarrow |0\rangle$

Dual-Rail Encoding



Use two orthogonal modes: information encoded in **which mode** contains the photon

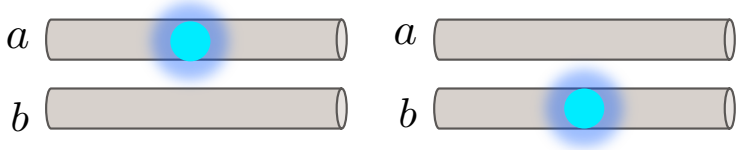
Challenge:

Requires two optical modes per qubit

Examples of dual rail photonic qubit encodings

Path qubit

$|0_L\rangle = |1\rangle_a |0\rangle_b \qquad |1_L\rangle = |0\rangle_a |1\rangle_b$



Photon polarisation

$|0_L\rangle = |H\rangle \qquad |1_L\rangle = |V\rangle$



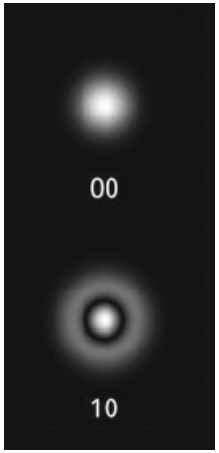
Horizontally polarised photon: $|H\rangle$

Vertically polarized photon: $|V\rangle$

Frequency bins

$|0_L\rangle = |\omega_1\rangle \qquad |1_L\rangle = |\omega_2\rangle$





Orbital angular momentum (OAM)

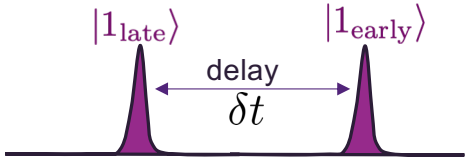
$|0_L\rangle = |1\rangle_{00}|0\rangle_{10}$

$|1_L\rangle = |0\rangle_{00}|1\rangle_{10}$

Spatial modes

Time bins

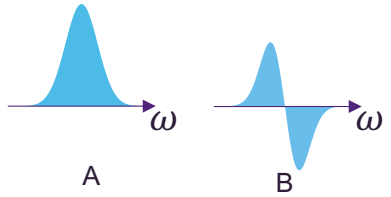
$|0_L\rangle = |1_{\text{early}}\rangle \quad |1_L\rangle = |1_{\text{late}}\rangle$



Pulse modes

$|0_L\rangle = |1\rangle_A|0\rangle_B$

$|1_L\rangle = |0\rangle_A|1\rangle_B$

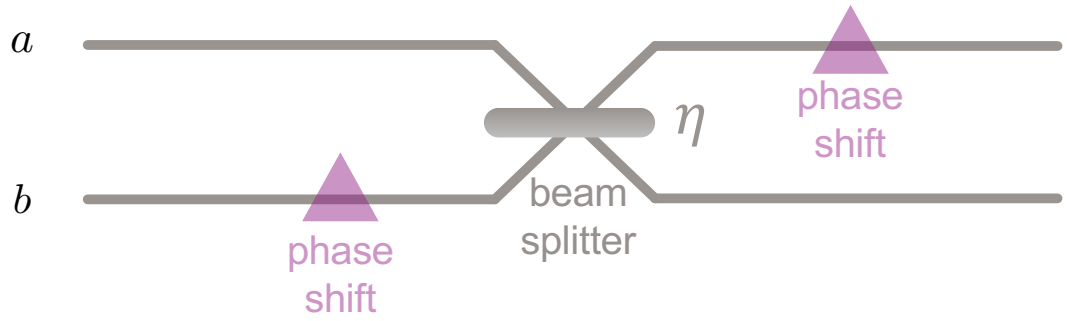


Time-frequency modes

Dual rail encoding

Spatial mode (path) qubit

$$|0_L\rangle = |1\rangle_a |0\rangle_b \qquad |1_L\rangle = |0\rangle_a |1\rangle_b$$



For a spatial mode qubit, any arbitrary single-qubit unitary can be realized as:

$$U = \mathrm{PS}(\phi_1)\,\mathrm{BS}(\eta)\,\mathrm{PS}(\phi_2)$$

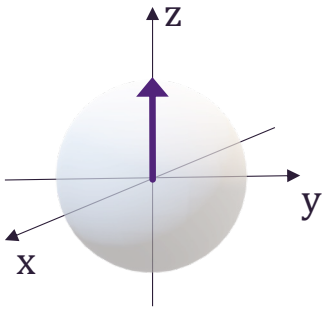
Photon polarisation qubit

Computational basis states



Horizontally polarised photon: $|H\rangle$

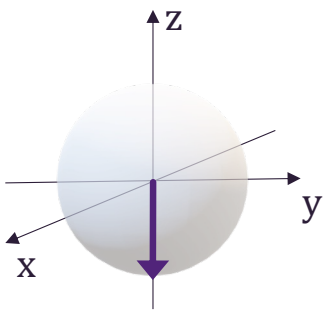
$$|H\rangle = |0\rangle$$





Vertically polarized photon: $|V\rangle$

$$|V\rangle = |1\rangle$$

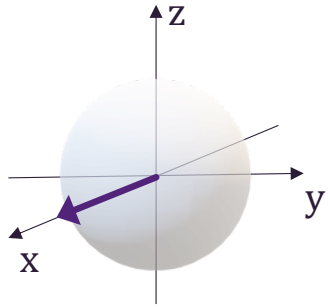


Qubits can be encoded in photon polarisation.

The computational basis states $\{|0\rangle, |1\rangle\}$ may be encoded in horizontal and vertical polarised photons respectively.

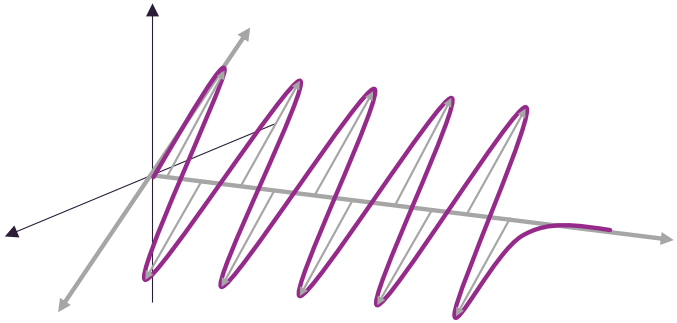
Even though the polarisation angle between horizontal and vertical is $90°$, $|H\rangle$ and $|V\rangle$ are orthogonal and antiparallel on the Bloch sphere.

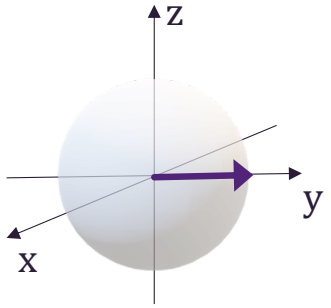
Photon polarisation qubit



$$|+\rangle = |\nearrow\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

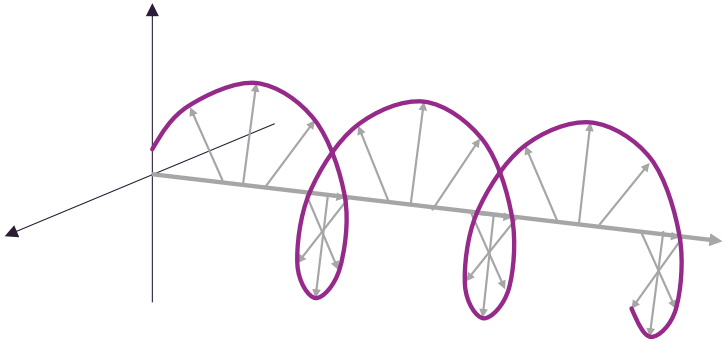
Diagonally polarised





$$|+i\rangle = |\circlearrowleft\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$$

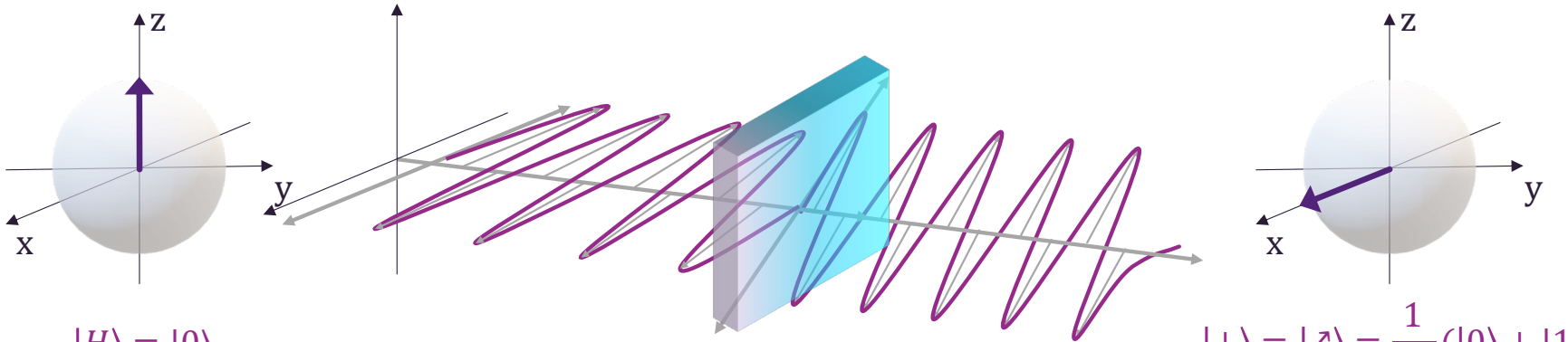
Circularly polarised



Photon polarisation – qubit operations

In order to implement single qubit unitary operations on polarisation encoded qubits, we need to manipulate the polarisation of light.

We can do this using wave plates. Waveplates are made from **birefringent materials** which have slightly different indices of refraction for light polarized in different orientations. They delay orthogonal polarization differently, so they recombine with a different phase.



$|H\rangle = |0\rangle$

Horizontally polarised

half wave plate

$|+\rangle = |\nearrow\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

Diagonally polarised

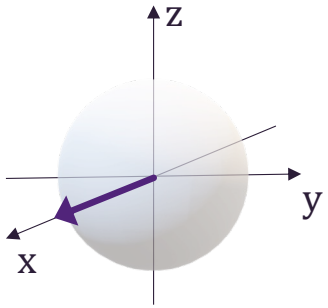
A half-waveplate applies a rotation of π about an axis in the xz plane (determined by the orientation of the HWP).

Photon polarisation – qubit operations

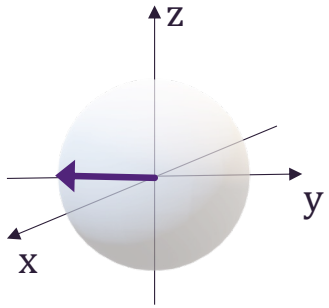
Quarter wave plate

A quarter-waveplate applies a rotation of $\pi/2$ about an axis in the xz plane (determined by the orientation of the QWP).

The quarter wave plate takes linearly polarised light to circularly polarised light.



$$U_{\mathrm{QWP}} = \begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix}$$

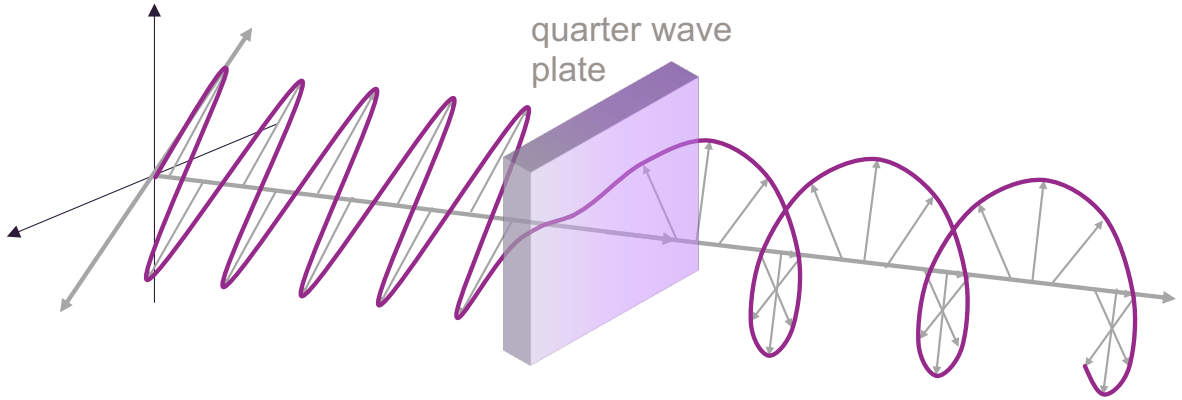


$$|+\rangle = |\nearrow\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Diagonally polarised

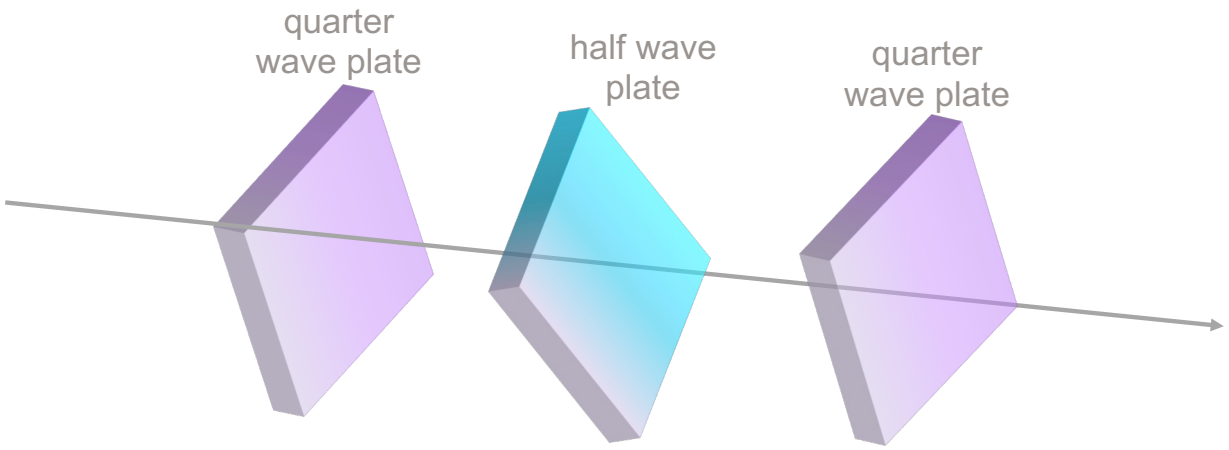
$$|-i\rangle = |\circlearrowright\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$$

Circularly polarised



Photon polarisation qubit

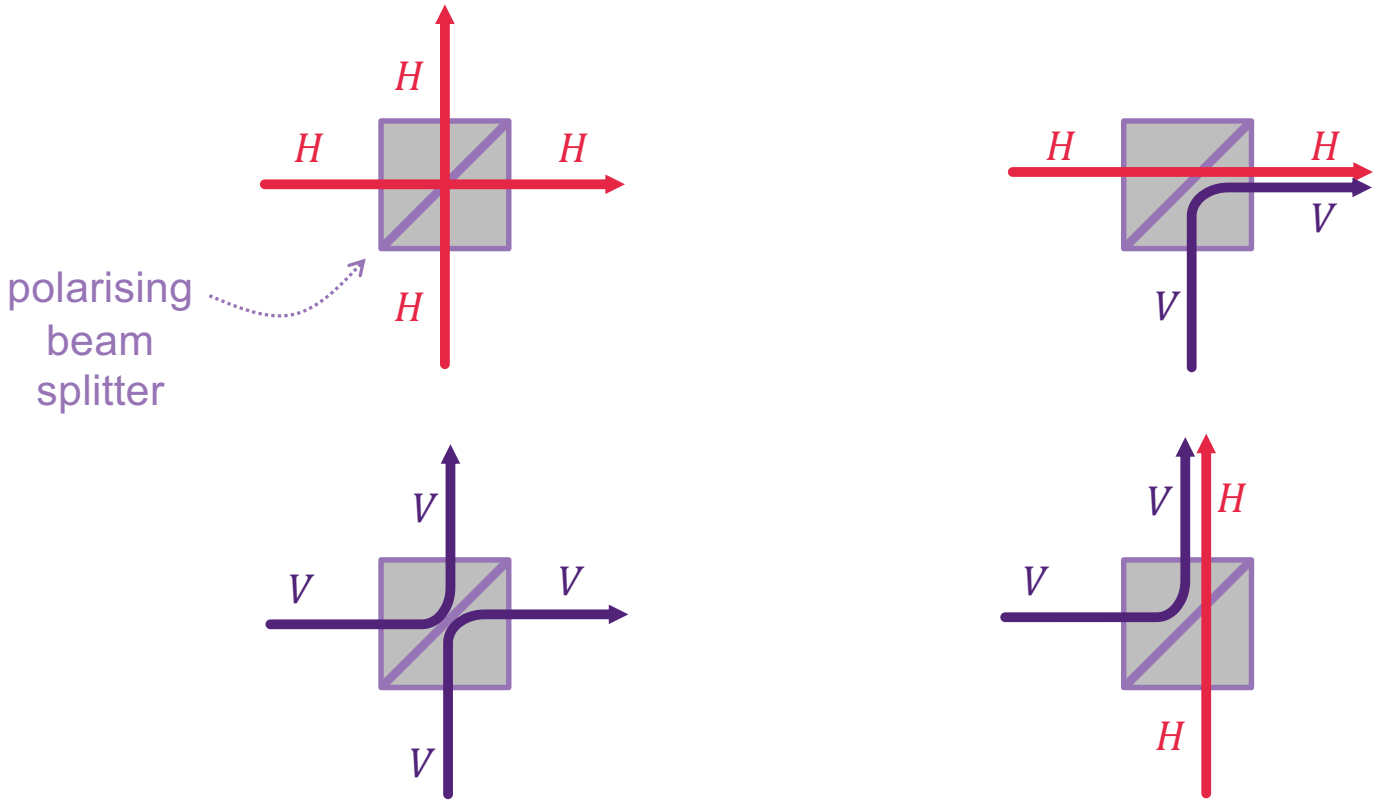
Arbitrary single qubit unitary



For a polarisation qubit, any arbitrary single-qubit unitary can be realized as:

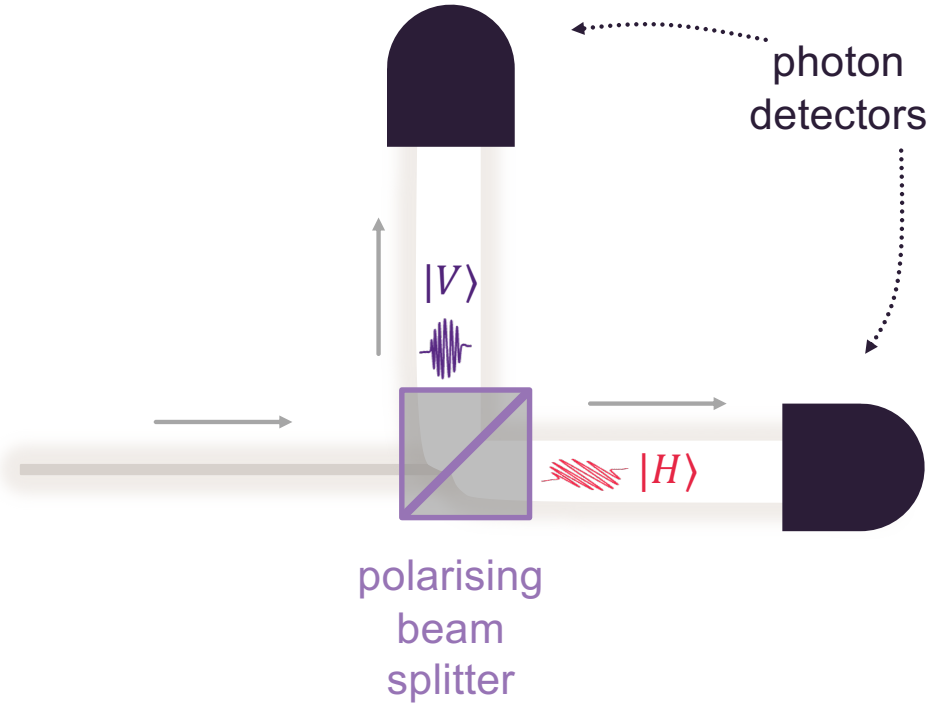
$$U = \mathrm{QWP}(\phi_1)\,\mathrm{HWP}(\phi_2)\,\mathrm{QWP}(\phi_3)$$

Polarising beam splitter (PBS)



- A polarising beam-splitter splits light into two different spatial modes each containing orthogonal polarisation states.
- For example, vertical polarised photons are reflected and horizontal polarised photons are transmitted

Photon polarisation - qubit measurement



- Photon detectors at the end of each output of the polarising beam splitter are needed to perform measurement of the qubit in the computational basis

- polarising beam splitter converts polarisation qubits to spatial mode qubits